MIL-T-12167E
19 October 1983
SUPERSEDING
MIL-T-12167D
5 April 1977

MILITARY SPECIFICATION

TIPS, NOZZLE, FIRE-HOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tips which, when attached to the outlet end of a shutoff nozzle, govern the rate of discharge of water or chemical.

1.2 Classification. The tips shall be the following classes, forms, and discharge opening sizes, as specified (see 6.2):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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Class A - For l-inch shutoff nozzle (see figure 1).

Form b - NH threads (3/4 inch).
Form C - NH threads (1 inch).

Discharge opening sizes:

1/4 inch
5/16 inch
3/8 inch
5/8 inch

Class B - For l-1/2-inch shutoff nozzle (see figure 1).

Form a - NH threads (1 inch).
Form b - NH threads (l-l/2-inch).

Discharge opening sizes:

3/8 inch
1/2 inch
5/8 inch

Class C - For 2-1/2-inch shutoff nozzle (see figure 2).

Form b - NH threads (1-1/2 inch).
Form C - NH threads (2-1/2 inch).

Discharge opening sizes:

5/8 inch
3/4 inch
7/8 inch
1 inch
1-1/8 inches
1-1/4 inches
1-3/8 inches
1-1/2 inches

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2,1.1 Specifications, and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the

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Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-775	- Packaging of Hose, Hose Assemblies, Rubber, Plastic, Fabric, or Metal (Including Tubing), and Fittings, Nozzles, and Strainers.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-sensitive.
MIL-P-23377	- primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28/10	- Screw Thread Standards for Federal Services, Section 10, American National Hose Couplings and Fire-Hose Coupling Threads.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 16	- Free-Cutting Brass Rod, Bar, and Shapes for Use in Screw Machines.
ASTM B 26	- Aluminum-Alloy Sand Castings.
ASTM B 124	- Copper and Copper-Alloy Forging Rod, Bar, and Shapes.
ASTM B 584	- Castings for General Applications, Specifications for Copper Alloy Sand.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The fire hose nozzle tips (hereinafter called “tips”) shall be fabricated of cast aluminum or brass, as specified (see 6.2), and shall be similar to the tips shown in figures 1 and 2 (see 6.5)

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Material deterioration and control. The tips shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of

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epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Cast aluminum. Aluminum castings shall conform to ASTM B 26, Alloy A03560, Condition T6, Alloy A17710, or Alloy A05350.

3.3.3 Brass.

3.3.3.1 Brass bars. Brass bars shall conform to ASTM B 16.

3.3.3.2 Drop forged brass. Drop forged brass shall conform to ASTM B 124, Alloy C37700.

3.3.3.3 Cast brass. Brass castings shall conform to ASTM B 584, Alloy C83600 or C84400.

3.4 Threads. All threads shall be American National Fire Hose Coupling threads (NH) conforming to FED-STD-H28/10.

3.5 Design. The waterway through the tips shall be conical in shape and shall terminate in a straight, smooth bore having a length of not less than 1-1/2 times the diameter of the respective discharge openings. The discharge end shall be counterbored to a diameter 1/8 inch larger than the discharge orifice for a depth of 1/16 inch.

3.6 Performance. The tips shall show no evidence of distortion when subjected to a hydrostatic pressure of 225 pounds per square inch gage (psig). The tips shall discharge a uniform, continuous stream of water at 150 psig nozzle pressure.

3.7 Gaskets. Each internal thread shall be furnished with a fitted gasket.

3.8 Identification marking. Tips shall be identified in accordance with MIL-STD-130. Each tip shall have the discharge opening size stamped or cast on the outside in figures not less than 1/4 inch in height.

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3.9 Class A, for l-inch shutoff nozzle. The class A tips shall be not less than 3-1/2 inches long. Discharge opening sizes shall be as specified (see 1.2 and 6.2) and shall be of the following thread forms .

3.9.1 Form b tips. The form b tips shall have 3/4-inch-8NH internal threads.

3.9.2 Form c tips. The form c tips shall have l-inch-8NH internal threads.

3.10 Class B, for l-1/2-inch shutoff nozzle. The class B tips shall be not less than 4-1/2 inches long. Discharge opening sizes shall be as specified (see 1.2 and 6.2) and shall be of the following thread forms.

3.10.1 Form a tips. The form a tips shall have l-inch-8NH internal threads.

3.10.2 Form b tips. The form b tips shall have l–1/2-inch-9NH internal threads.

3.11 Class C, for 2-1/2-inch shutoff nozzle. The class C tips shall be not less than 4-1/2 inches long. Discharge opening sizes shall be as specified (see 1.2 and 6.2) and shall be of the following thread forms.

3.11.1 Form b tips. The form b tips shall have l-1/2-inch-9NH internal threads.

3.11.2 Fox-m c tips. The form c tips shall have 2–1/2-inch-7-l/2NH internal threads.

3.12 Workmanship. All parts of the tips shall be clean and free from sand, dirt, fins, pits, sprues, scale, and other harmful material. All edges shall be rounded or chamfered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article tips shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article tips shall be tested as specified in 4.5.2. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot. For the purpose of inspection, a lot shall consist of all tips of the same class, form, and size offered for delivery at one time.

4.4.2 Sampling.

4.4.2.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.4.2.2 Tests. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.3 Examination. Samples selected in accordance with 4.4.2.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.4 Tests. Samples selected in accordance with 4.4.2.2 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The tips shall be examined as specified herein for the following defects:

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Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Class, form, and size not as specified.
106. Finish plan not adequate.
107. Design not as specified.
108. Threads not as specified or damaged.
109. Dimensions not as specified.
110. Defective castings such as blowholes, porosity, cracks, or mismatching.
111. Edge and surface defects such as gouges, nicks, or cracks.

Minor

201. Gaskets missing or size not as specified.
202. Finish not as specified.
203. Identification and other marking missing or illegible.

4.5.2 Tests.

4.5.2.1 Hydrostatic pressure . Hydrostatic pressure tests shall be conducted according to standard methods using commercial or equivalent hydrostatic testing apparatus . The tips shall be subjected to a pressure of 225 psig, and then examined . Evidence of any distortion shall constitute failure of this test.

4.5.2.2 Discharge. The tips shall be subjected to a discharge test at 150 psig nozzle pressure conducted according to standard methods using commerical or equivalent hydrostatic indication apparatus to measure static pressure head at the inlet of the nozzle. Failure to discharge a uniform, continuous stream of water at 150 psig nozzle pressure shall constitute failure of this test.

4.6 Inspection of packaging. Preservation, packing, and marking for level A or B shall be inspected in accordance with the Quality Assurance Provisions of MIL-P-775. Commercial preservation, packing, and marking shall be examined to determine compliance with MIL-STD-1188.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The tips shall be preserved in accordance with the level A preservation and packaging requirements of MIL-P-775.

5.1.2 Commercial. The tips shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The tips, preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-P-775.

5.2.2 Level B. The tips, preserved as specified in 5.1, shall be packed in accordance with the level B packing requirements of MIL-P-775.

5.2.3 Commercial. The tips, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL–STD-1188.

6. NOTES

6.1 Intended use. The tips are intended for use on corresponding sizes of fire hose, shutoff nozzles, and playpipes used in combating fires.

6.2 Ordering data, Acquisition documents shall specify the following:

a. Title, number , and date of this specification.
b. Class, form, and discharge opening size required (see 1.2, 3,9, 3.10 and 3.11).
c. When a first article is required for inspection and approval , and number of tips required (see 3.2).
d. Material required (see 3.1).
e. Degree of packing required (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction tip. The preproduction tip should consist of one or more tips. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, texts, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Information figures. Figures 1 and 2 show tips which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other tips which conform to this specification.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Air Force - 84
DLA - CS

User activities:
Navy - MC, AS

Preparing activity:
Army-ME

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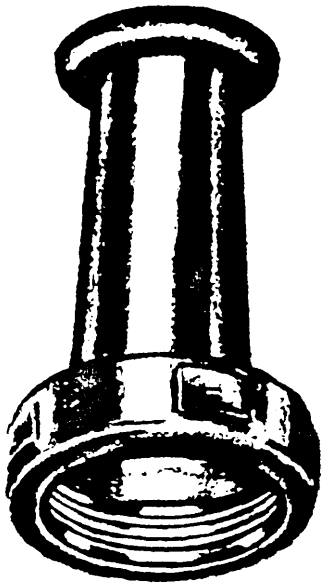


FIGURE 1. Tip, nozzle, fire hose, class A & B.

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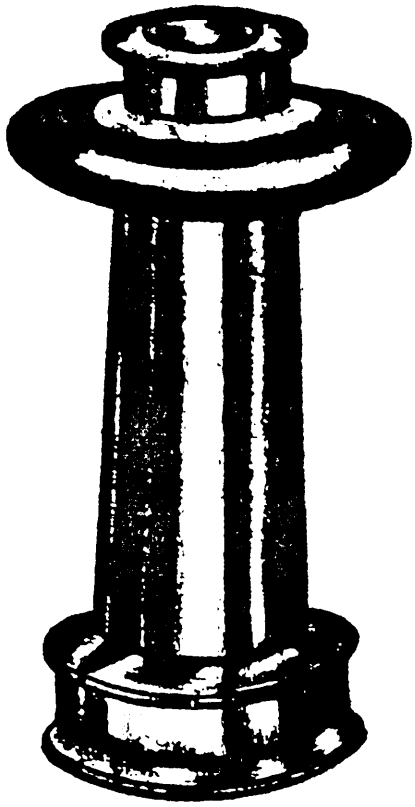


FIGURE 2. Tip, nozzle, fire hose, class C.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-12167E	Tips, Nozzle, Fire-Hose

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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